# Lecture 14: 4-vectors continued

Based on Morin's *Introduction to Classical Mechanics*, Ch. 13.4 –13.6

## Outline

1. Recap lecture 13
2. Energy and Momentum
3. Force and Acceleration
4. Physical Laws

## 1 Recap lecture 13

In Lecture 13, we defined 4-vectors. We explained their properties and provided examples. In SR, the most important 4-vectors are the spacetime interval $dS = (dt, dx, dy, dz)$, the 4-velocity $V = (\gamma, \gamma\mathbf{v})$ and the 4-momentum $P \equiv mV = (m\gamma, m\gamma\mathbf{v}) = (E, \mathbf{p})$. By starting with the spacetime interval and using successive derivatives on the spacetime interval $dS$ with respect to proper time $d\tau$, we can construct an infinite tower of 4-vectors. Taking derivatives with respect to $\tau$ or multiplying by $m$ (as is done for the 4-momentum) results in a new 4-vector, as both $d\tau$ and $m$ are frame-independent quantities (and hence should not be affected by a Lorentz transformation).

Using 4-vectors, you find that there is a natural definition of relativistic energy and (3) momentum, as the time and spatial components of the 4-momentum vector. While we used heuristic arguments to build these quantities in Lecture 11, we obtained them here by simple manipulation of the spacetime interval 4-vector.

Similarly, by taking yet another derivative, either from $V$ or $P$, with respect to $\tau$, we found that you can find an expression for the 4-acceleration and the 4-force, respectively. The spatial components of the 4-force are equivalent to the force we derived in lecture 12, showing consistent results. Furthermore, the 4-force $F$ can also be related to the 4-acceleration $A$ as $F = mA$, which again shows consistency with our idea of Newton's notion of forces.

We also introduced a bit of tensor notation, e.g.

$$dS^2 = \eta_{\mu\nu} dx^\mu dx^\nu. \tag{1}$$

Combined with the tensor notation for Lorentz transformations, this allowed us to find some formal proofs of the nature and properties of 4-vectors. We do not expect you to fully comprehend the tensor algebra presented in this course. It serves as a complementary component to the details shown in Morin's book.

## 2 Energy and Momentum

Since $P$ is a 4-vector and we argued the norm of a 4-vector is frame-independent, you could move to a frame where an object is at rest ($P(v \neq 0) \rightarrow P'(v' = 0)$). In this frame, we have $P' = (m, 0, 0, 0)$, and hence $P'^2 = m^2 = P^2$. We already knew this, of course, but it is nice to see how this could have been derived very easily by a simple transformation.

If you have a collection of particles, knowledge of the norm is useful. Suppose you have a collection of particles $n$, then for any subset of particles $m < n$, we know

$$\left(\sum_{i=1}^{m} E\right)^2 - \left(\sum_{i=1}^{m} \mathbf{p}\right)^2 = \text{invariant}. \tag{2}$$

This is true because it presents the squared norm of the sum of the energy-momentum 4-vectors of the subset $m$ of $n$ particles. As we had shown in lecture 13, a linear combination of 4-vectors is again a 4-vector.

In the center of mass (CM) frame (lecture 9) $\sum \mathbf{p} = 0$, the above would reduce to the square of the energies, which reduces to $m^2$ for a single particle.

Since $P$ is a 4-vector, according to the definition of a 4-vector, its components should transform according to the Lorentz transformations, i.e., (motion $\beta$ in the $x$ direction)

$$\begin{aligned} E &= \gamma(E' + \beta p'_x), \\ p_x &= \gamma(p'_x + \beta E'), \\ p_y &= p'_y, \\ p_z &= p'_z. \end{aligned} \tag{3}$$

This is equivalent to the transformation rules we derived in lectures 10 and 11.

Since $E$ and $\mathbf{p}$ are part of the same 4-vector, if one of the two is conserved in every frame in a collision, then so is the other. To see this, you can consider the 4-momentum change 4-vector $\Delta P \equiv P_{\text{after}} - P_{\text{before}}$. If, e.g., $E$ is conserved in every frame, then $\Delta P_0 = 0$ in every frame. By virtue of one of the theorems discussed in the previous lecture, we know that the other component should be zero, and hence $\mathbf{p}$ should also be conserved in every frame.

## 3 Force and Acceleration

We will assume $m$ is constant and does not depend on time.

$$P \cdot P = E^2 - \mathbf{p}^2 = m^2. \tag{4}$$

Differentiating with respect to coordinate time gives

$$2E\frac{dE}{dt} - 2\mathbf{p} \cdot \frac{d\mathbf{p}}{dt} = 0. \tag{5}$$

Therefore,

$$\frac{dE}{dt} = \frac{\mathbf{p}}{E} \cdot \mathbf{F} = \mathbf{v} \cdot \mathbf{F}, \tag{6}$$

where $\mathbf{F} = d\mathbf{p}/dt$ and $\mathbf{p}/E = \mathbf{v}$. Hence

$$F = \frac{dP}{d\tau} = \gamma \left( \frac{dE}{dt}, \mathbf{F} \right) = \gamma \left( \mathbf{v} \cdot \mathbf{F}, \mathbf{F} \right). \tag{7}$$

This is a derived relation, not an additional assumption. In the instantaneous inertial frame $S'$, $\mathbf{v}' = 0$, so $dE'/dt' = 0$ and therefore

$$F' = (0, \mathbf{F}'). \tag{8}$$

Next, consider another frame $S$, in which the particle moves with velocity $v$ $(= \beta)$ in the $x$ direction. Since $F'$ is a 4-vector, it transforms according to the Lorentz transformations:

$$\begin{aligned} F_0 &= \gamma(F'_0 + \beta F'_1) = \gamma \beta F'_x, \\ F_1 &= \gamma(F'_1 + \beta F'_0) = \gamma F'_x, \\ F_2 &= F'_2 = F'_y, \\ F_3 &= F'_3 = F'_z. \end{aligned} \tag{9}$$

In addition, we also know that

$$\begin{aligned} F_0 &= \gamma \frac{dE}{dt}, \\ F_1 &= \gamma F_x, \\ F_2 &= \gamma F_y, \\ F_3 &= \gamma F_z. \end{aligned} \tag{10}$$

We can now equate the two, to find

$$\begin{aligned} \frac{dE}{dt} &= \beta F'_x, \\ F_x &= F'_x, \\ F_y &= F'_y/\gamma, \\ F_z &= F'_z/\gamma. \end{aligned} \tag{11}$$

This is the same as we found in lecture 12, when we showed that the force in the longitudinal direction $(x)$ remains the same, while the force along the transverse direction $(y, z)$ is amplified by a factor $\gamma$ in the particle frame $S'$.

**Why do longitudinal and transverse forces produce different accelerations?** At any event, *longitudinal* means parallel to the particle's instantaneous velocity, and *transverse* means perpendicular to it. Choose the $x$ axis along the instantaneous velocity, so that $\mathbf{v} = (v, 0, 0)$. The Lorentz factor depends on the *speed* rather than on the direction of motion, and therefore

$$\frac{d\gamma}{dt} = \gamma^3 \mathbf{v} \cdot \mathbf{a} = \gamma^3 v a_x. \tag{12}$$

Thus a longitudinal acceleration changes the speed and hence $\gamma$, whereas a transverse acceleration only turns the velocity vector to first order. Differentiating the momentum components makes the distinction explicit:

$$\begin{aligned} F_x &= \frac{d(\gamma m v_x)}{dt} = m\gamma^3 a_x, \\ F_y &= \frac{d(\gamma m v_y)}{dt} = m\gamma a_y, \end{aligned} \tag{13}$$

where the term involving $d\gamma/dt$ in the second line vanishes at this event because $v_y = 0$. This is the physical reason that a given transverse force produces a larger coordinate acceleration than an equally large longitudinal force when $\gamma > 1$.

The ratio of $F_y/F_x = \gamma^{-1}F'_y/F'_x$, i.e. the ratio of the forces decreases by a factor $\gamma$ when going from $S'$ to $S$ (hence the norm of the vector $(F_x, F_y)$ will decrease).

The first line combined with the second line tells us that $dE = F_x dx$, which is the work-energy result we derived in lecture 12 and is analogous to the result for classical motion in lecture 9.

Note that we can not write $F'_y = F_y/\gamma$ (by switching the two frames). This is because when considering forces on the particle, there is one specific frame, the particle frame ($S'$), where the forces apply, which is part of the underlying assumption when the 4-vectors were derived.

We can apply the same analysis for the acceleration 4-vector. In frame $S'$, the instantaneous inertial frame of the particle, we have

$$A' = (0, \mathbf{a}'), \tag{14}$$

because $v' = 0$ in $S'$. In the frame $S$, where the particle is moving with velocity $v = \beta$ in the $x$ direction, we can relate the components of $A$ to those of $A'$ using the Lorentz transformations:

$$\begin{aligned} A_0 &= \gamma(A'_0 + \beta A'_1) = \gamma\beta a'_x, \\ A_1 &= \gamma(A'_1 + \beta A'_0) = \gamma a'_x, \\ A_2 &= A'_2 = a'_y, \\ A_3 &= A'_3 = a'_z. \end{aligned} \tag{15}$$

From the expression for the acceleration 4-vector derived in lecture 13, we also have

$$\begin{aligned} A_0 &= \gamma^4 v a_x, \\ A_1 &= \gamma^4 a_x, \\ A_2 &= \gamma^2 a_y, \\ A_3 &= \gamma^2 a_z. \end{aligned} \tag{16}$$

Equating the two again, yields:

$$\begin{aligned} a_x &= a'_x/\gamma^3, \\ a_y &= a'_y/\gamma^2, \\ a_z &= a'_z/\gamma^2. \end{aligned} \tag{17}$$

We recover the same results found in Lecture 12. In contrast to the force transformation, we find $a_y/a_x = \gamma a'_y/a'_x$. This means that $\mathbf{F} = m\mathbf{a}$ does not generally apply in SR. If it were true in one frame, it would not be true in another.

### Example 1: Acceleration for circular motion

Consider a particle moving around a circle $x^2 + y^2 = r^2$ ($z = 0$) with constant speed as observed in a lab frame $S$.[1] At the instant the particle crosses the negative $y$ axis, find

[1] The local relation $d\tau = \sqrt{1 - v^2}\, dt$ is valid for any timelike worldline. Here the speed $v$ is constant, although the velocity vector $\mathbf{v}$ changes direction.

the three-acceleration and four-acceleration in $S$ and in the instantaneous inertial frame comoving with the particle.

First, we write for the radial position of the particle at some instant

$$\mathbf{r} = r(\hat{x}\cos\omega t + \hat{y}\sin\omega t), \tag{18}$$

with $\omega$ the angular velocity. To get the velocity, we differentiate with respect to $t$:

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \omega r(-\hat{x}\sin\omega t + \hat{y}\cos\omega t). \tag{19}$$

Hence we have $|\mathbf{v}| = v = \omega r$. To obtain the acceleration, we differentiate again:

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = -\omega^2 r(\hat{x}\cos\omega t + \hat{y}\sin\omega t). \tag{20}$$

Thus we find $|\mathbf{a}| = a = v^2/r$. In the moment that the particle is crossing the negative $y$ axis, the acceleration is zero in the $x$ direction, and all the acceleration is in the positive $y$ direction. With $\Theta = \omega t$, we have $\Theta = \frac{3\pi}{2}$ and hence

$$\mathbf{a} = (0, v^2/r, 0). \tag{21}$$

Substituting $\mathbf{a} = (0, v^2/r, 0)$ into Eq. (16), we find

$$A = (0, 0, \gamma^2 v^2/r, 0). \tag{22}$$

To obtain the components in the $S'$ frame, we use that the components of $A$ are related to the components of $A'$ through a Lorentz transformation in the $x$ direction. This means that the $A_2 = A'_2$ (and all the other components that are zero remain zero). Thus, we have:

$$A' = (0, 0, \gamma^2 v^2/r, 0) \tag{23}$$

The 3 acceleration in $S'$, where $v' = 0$ and $\gamma' = 1$, should be related to the spatial components of $A'$:

$$\mathbf{a}' = (0, \gamma^2 v^2/r, 0). \tag{24}$$

This also follows from Eq. (17): since $a_y = v^2/r$, it gives $a'_y = \gamma^2 v^2/r$.

**What is the physical point?** Constant speed does not mean zero acceleration: the velocity changes direction, just as in Newtonian circular motion. Changing frame does not create an additional physical acceleration; we are describing the same motion using different coordinate times and velocities. For this boost along $x$, the transverse four-acceleration component is unchanged, $A'_2 = A_2$, but the ordinary three-accelerations are different:

$$a_y = \frac{v^2}{r}, \qquad A_2 = \gamma^2 a_y = A'_2 = a'_y, \qquad \boxed{a'_y = \gamma^2 a_y.}$$

The last equality uses $\gamma' = 1$ in the instantaneous rest frame. An accelerometer carried by the particle measures the *proper acceleration*, namely the magnitude of the three-acceleration in that frame:

$$\alpha = |\mathbf{a}'| = \gamma^2 \frac{v^2}{r}.$$

It does not measure the lab-frame coordinate acceleration $|\mathbf{a}| = v^2/r$. These values agree in the Newtonian limit $\gamma \simeq 1$.

**An intuitive explanation of the factor $\gamma^2$.** Keep $S'$ as one fixed inertial frame, chosen to be comoving at the event under discussion. Over a short interval near that event, the transverse displacement is unchanged by the boost, $\Delta y' = \Delta y$, while along the particle's motion $\Delta t' \simeq \Delta t/\gamma$ to leading order. Since the initial transverse velocity is zero in both frames,

$$\Delta y \simeq \frac{1}{2} a_y (\Delta t)^2, \qquad \Delta y' \simeq \frac{1}{2} a'_y (\Delta t')^2.$$

The same transverse displacement over the shorter coordinate-time interval therefore gives $a'_y = \gamma^2 a_y$ in the limit of vanishing interval. This argument is local; it does not treat $S'$ as a frame that follows the particle throughout its circular orbit.

Finally, *momentarily at rest does not mean unaccelerated*. The particle has $\mathbf{v}' = 0$ at this event, but its velocity in $S'$ is changing, so $\mathbf{a}' \neq 0$.

# 4 The form of physical laws

One of the key postulates of SR is that all inertial frames are equivalent. If a physical law holds in one inertial frame, it will hold in all frames. If not, we would be able to differentiate between frames. For example, we found that $\mathbf{F} = m\mathbf{a}$ can not be a physical law (although we long defended it to be!) since $\mathbf{F}$ transforms differently from $\mathbf{a}$ as we showed in the previous section. It might be true in one frame, but it certainly can not be true in all frames.

As a guiding principle, for a statement or physical law to hold in all frames, it should involve 4-vectors (as these transform according to the Lorentz transformations). Suppose in one frame we have a statement that relates two 4-vectors, $A = B$. Does this hold in all inertial frames? Yes, since we can apply the Lorentz transformation (and an infinite series if we absolutely want to) and find

$$A = B \rightarrow \Lambda A = \Lambda B \rightarrow A' = B', \tag{25}$$

where we have again used $\Lambda$ as our Lorentz transformation matrix. This relation therefore also holds in a new frame $S'$. Only relations that are physically and mathematically valid in one frame are valid in all frames. For example, $P = mV$ and the four-vector equation $F = mA$ are covariant relations, whereas the three-vector equation $\mathbf{F} = m\mathbf{a}$ is not generally valid in SR.

We have also seen that scalars are frame independent, so physical laws may contain scalars such as $m$, or may be scalar equations such as $P \cdot P = m^2$. If they are true in one frame, they hold in all frames. In SR, we concern ourselves with scalars and four-vectors. Physical laws may also involve higher-rank tensors: scalars are rank-zero tensors, while four-vectors are rank-one tensors. Such higher-rank tensors can be built from four-vectors.

There is nothing really new here. In Newtonian mechanics, we have $\mathbf{F} = m\mathbf{a}$ as a possible physical law because both sides are 3-vectors. $\mathbf{F} = m(2a_x, a_y, a_z)$ is not a possible law because $(2a_x, a_y, a_z)$ is not a 3-vector, since it depends on which axis you label the $x$ axis. It makes no sense for a physical law to depend on your arbitrary choice of coordinates.